

Interpretable Multi-Source Machine Learning for Urban Classification in Data-Scarce Regions

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Abstract

Rapid urban expansion in data-scarce regions demands accurate and interpretable methods for monitoring urban growth. This study presents an interpretable multi-source machine learning framework for urban classification in Lagos, Nigeria, integrating spectral, socio-economic, and temporal geospatial data. A dataset of 36,000 samples spanning 2000–2024 was constructed using spectral bands, derived indices, night-time lights, population density, and elevation. We evaluate Random Forest, XGBoost, and a multilayer perceptron, achieving consistent performance across models (accuracy = 0.87, AUC > 0.94). Notably, results show that feature integration plays a more critical role than model complexity, with ensemble methods performing comparably to neural networks. To enhance interpretability, SHAP-based explainable AI is applied to quantify feature contributions. The analysis identifies night-time lights and population density as dominant drivers of urban classification, while spectral variables provide complementary discriminatory power. These findings demonstrate that combining heterogeneous data sources with interpretable machine learning enables both accurate prediction and meaningful insight. Overall, the proposed framework provides a scalable and transparent approach for urban monitoring in rapidly developing, data-limited environments.

1 Introduction

Urbanization is one of the most significant forms of land transformation occurring globally, with profound implications for environmental sustainability, infrastructure planning, and socio-economic development. Rapid urban growth in developing regions, particularly in sub-Saharan Africa, has intensified the need for accurate, scalable, and interpretable methods for monitoring urban expansion [Seto *et al.*, 2012; United Nations Human Settlements Programme, 2022]. Lagos, Nigeria, represents one of the fastest growing megacities in Africa, experiencing substantial population growth and spatial expansion over recent decades [United Nations Human Settlements Programme, 2022]. Understanding these

dynamics requires robust approaches capable of integrating diverse data sources and capturing complex urban patterns.

Remote sensing has become an indispensable tool for mapping urban land cover due to its ability to provide consistent spatial observations across large geographic areas. Traditional classification approaches based on spectral thresholds or statistical classifiers have been widely applied [Yu and Fang, 2023; Weng, 2012]. However, urban environments are inherently heterogeneous, often consisting of mixtures of built-up surfaces, vegetation, and bare soil, which limits the effectiveness of conventional methods.

Machine learning methods have significantly improved classification performance by modeling nonlinear relationships in geospatial data. Ensemble approaches such as Random Forest are widely used due to their robustness and ability to handle high-dimensional features [Belgiu and Drăguț, 2016]. Gradient boosting methods, particularly XGBoost, further enhance predictive performance through sequential optimization [Chen and Guestrin, 2016]. In parallel, deep learning models have demonstrated strong capability in capturing complex feature interactions and extracting patterns from large datasets [L. Ma and Johnson, 2019].

Recent studies have emphasized the integration of multi-source geospatial datasets with machine learning to improve urban mapping and classification performance [Sun *et al.*, 2023]. In particular, spectral information, night-time lights, and population data provide complementary information for distinguishing urban and non-urban areas. Night-time lights have been shown to correlate strongly with urbanization and economic activity, while population density captures patterns of human settlement [Li and Zhou, 2016; Tatem, 2017]. These heterogeneous data sources enable more comprehensive modeling of urban dynamics

Despite advances in predictive modeling, interpretability remains a key challenge in applying machine learning to urban studies. Explainable Artificial Intelligence (XAI) techniques provide a principled approach for understanding model predictions and have become increasingly important across remote sensing and geospatial applications [Roussel and Böhm, 2023; Höhl *et al.*, 2024]. Among these, SHapley Additive exPlanations (SHAP) offers a unified framework for quantifying feature contributions based on cooperative game theory. [Lundberg and Lee., 2017]

However, relatively few studies have combined multi-

source geospatial data, multiple machine learning paradigms, and explainable AI within a unified framework for urban classification, particularly in rapidly growing African cities. Addressing this gap is essential for improving both predictive accuracy and interpretability in urban monitoring.

To address these challenges, this study proposes an interpretable machine learning framework for urban classification in Lagos, Nigeria using multisource geospatial data. The approach integrates ensemble learning methods, deep learning, and SHAP-based explainable AI to achieve both high predictive performance and meaningful interpretation of model outputs.

While prior studies have explored machine learning for urban classification, this work advances the field by explicitly linking predictive performance with interpretable insights in a unified multi-source framework.

Contributions. This paper makes the following contributions:

- We propose an interpretable multi-source learning framework that integrates heterogeneous geospatial, socio-economic, and temporal data for urban classification in data-scarce environments.
- We show that, in multisource geospatial settings, feature representation and data integration are more critical than model complexity, with ensemble methods performing comparably to neural networks.
- We provide a comparative analysis of Random Forest, XGBoost, and neural networks on tabular geospatial data, offering empirical evidence of their effectiveness for urban classification.
- We apply SHAP-based explainable AI to identify the key drivers of urbanization, revealing consistent relationships between night-time lights, population density, spectral features, and urban patterns.
- We demonstrate that combining predictive modeling with explainability enables a transparent and scalable decision-support framework for urban monitoring in data-limited regions.

2 Materials and Methods

2.1 Dataset Preparation

A multisource geospatial dataset was developed by integrating optical, socioeconomic, and topographic variables derived from publicly available Earth observation datasets. Annual predictor layers were generated in Google Earth Engine (GEE) for six temporal snapshots (2000, 2005, 2010, 2015, 2020, and 2024) using cloud-masked median composites of Landsat Collection 2 Level-2 Surface Reflectance imagery. Urban reference labels were derived from the Global Human Settlement Layer (GHSL) built-up surface dataset by classifying pixels with non-zero built-up values as urban. Stratified random sampling was then used to extract balanced urban and non-urban observations at 30 m spatial resolution, producing approximately 6,000 labelled samples per year (3,000 per class) and a combined dataset of 36,000 observations.

Predictor variables were derived from a combination of satellite imagery, socio-economic datasets, and topographic

information. These variables included spectral bands such as Blue, Green, Red, Near Infrared (NIR), and Shortwave Infrared (SWIR), as well as derived indices including the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built-up Index (NDBI). In addition, night-time lights intensity and population density were incorporated to capture human activity patterns associated with urban development, while elevation was included to account for topographic influences. A temporal variable (Year) was also included to represent changes across time.

Spectral indices such as NDVI and NDBI were used to differentiate vegetation and built-up areas, consistent with established approaches in remote sensing-based urban mapping [Weng, 2012; Y. Zha and Ni, 2003]. Night-time lights and population datasets provide complementary information on anthropogenic activity and have been widely used as proxies for urbanization and economic activity [Tatem, 2017].

Prior to model training, the dataset was randomly split into training and testing subsets using an 80:20 ratio. This ensured that model performance could be evaluated on unseen data while maintaining sufficient samples for training. All analyses were implemented in Python using standard machine learning libraries in a Google Colab environment to ensure reproducibility. Missing population values were imputed using the median, while logarithmic transformation was applied to the Population and NightLights variables before model training to reduce skewness.

2.2 Machine Learning Models

Three classification algorithms were implemented using Python in a Google Colab environment to evaluate their performance in urban classification. The first model, Random Forest, is an ensemble learning method that constructs multiple decision trees and aggregates their predictions to improve classification accuracy and reduce overfitting [Z. Sun and Liang., 2024]. The second model, Extreme Gradient Boosting (XGBoost), is a gradient boosting algorithm that sequentially builds decision trees to minimize prediction error and has demonstrated strong predictive performance in a wide range of geospatial applications [Chen and Guestrin, 2016]. The third model is a multilayer perceptron (MLP) implemented in PyTorch. Unlike the tree-based ensemble models, the MLP learns hierarchical nonlinear feature representations through stacked fully connected layers, enabling a comparative assessment of deep learning on tabular multisource geospatial data. [T. Man and Ignatov, 2024]

The Random Forest classifier was implemented using 500 decision trees with bootstrap aggregation and parallel processing. XGBoost was configured with 500 estimators, a maximum tree depth of 6, a learning rate of 0.05, and row and feature subsampling ratios of 0.8 to improve generalization. The neural network was implemented as a multilayer perceptron comprising two fully connected hidden layers (64 and 32 neurons), each followed by batch normalization and dropout (0.3). The network was trained for 30 epochs using the Adam optimizer and binary cross-entropy loss with logits. These configurations were selected to provide a consistent comparison between ensemble learning and deep learning approaches while minimizing overfitting.

2.3 Model Evaluation

Model performance was evaluated using Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC). Accuracy provides an overall measure of correct classifications, whereas Precision, Recall, and F1-score offer complementary insights into classification performance, particularly for binary classification problems. AUC evaluates the model's ability to discriminate between urban and non-urban classes across different classification thresholds. Together, these metrics provide a comprehensive assessment of predictive performance and have been widely adopted for evaluating machine learning classifiers in remote sensing and pattern recognition studies [Fawcett, 2005; Sokolova and Lapalme, 2009].

2.4 Explainable AI Analysis

To enhance model interpretability, SHapley Additive exPlanations (SHAP) were applied to quantify the contribution of each predictor variable to model predictions. SHAP is based on cooperative game theory and provides consistent and locally accurate feature attributions for machine learning models [Lundberg and Lee., 2017]. This approach enables both global and local interpretation of model behavior, allowing for a detailed understanding of the factors influencing classification outcomes. Several SHAP-based visualizations were generated to support interpretation, including feature importance bar plots, beeswarm summary plots, and dependence plots. These visualizations provide insights into both the relative importance of predictor variables and the direction of their influence on urban classification, thereby helping to identify the key drivers of urban development [A. V. Ponce-Bobadilla and Stodtmann, 2024].

3 Results

3.1 Model Performance

Table 1 summarizes the predictive performance of the three evaluated classification models. Random Forest and XGBoost achieved identical Accuracy, Precision, Recall, and F1-score values of 0.87, demonstrating consistently strong performance in distinguishing urban and non-urban areas. XGBoost recorded the highest AUC (0.943), indicating a marginally better ability to discriminate between the two classes across different decision thresholds, while Random Forest achieved a nearly identical AUC (0.942). The multilayer perceptron (MLP) also demonstrated competitive performance but produced slightly lower Accuracy (0.86), F1-score (0.86), and AUC (0.941). Overall, the results indicate that tree-based ensemble methods provide highly effective and robust performance for multisource urban classification.

Model	Accuracy	Precision	Recall	F1-score	AUC
Random Forest	0.87	0.87	0.87	0.87	0.942
XGBoost	0.87	0.87	0.87	0.87	0.943
Neural Network	0.86	0.87	0.86	0.86	0.941

Table 1: Performance comparison of the evaluated classification models.

The Receiver Operating Characteristic (ROC) curves further illustrate the predictive performance of the evaluated models (Figure 1). XGBoost achieved the highest Area Under the Curve (AUC) value of 0.943, followed closely by Random Forest (0.942), while the multilayer perceptron (MLP) achieved an AUC of 0.941. The consistently high AUC values across all three models demonstrate strong discriminative ability for distinguishing urban from non-urban classes, with only marginal differences in predictive performance.

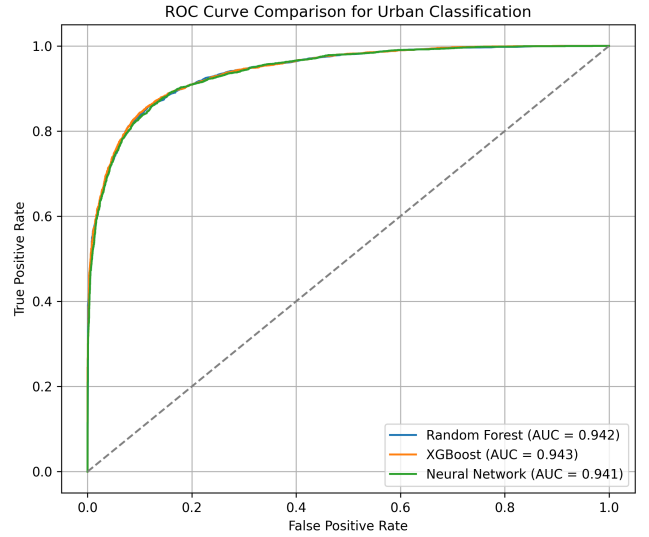


Figure 1: ROC curves comparing Random Forest, XGBoost, and neural network models, showing consistently high discriminative performance ($AUC > 0.94$), with XGBoost achieving the highest AUC.

The comparable performance across Random Forest, XGBoost, and the neural network suggests that feature representation plays a more critical role than model complexity in multisource geospatial classification tasks. This indicates that when heterogeneous data sources are well integrated, simpler ensemble methods can achieve performance comparable to more complex deep learning models.

3.2 Feature Importance

The relative importance of the predictor variables derived from the Random Forest model is presented in Figure 2. The Green spectral band was identified as the most influential predictor, followed by Night-time Lights and Population Density. The prominence of the Green band suggests that spectral reflectance provides strong discriminatory information for separating urban from non-urban land cover, while the importance of Night-time Lights and Population Density highlights the complementary contribution of human activity indicators. These findings demonstrate that integrating spectral, socioeconomic, and environmental variables improves urban classification.

3.3 SHAP Explainable AI Results

To further interpret the model predictions, SHAP analysis was conducted. The SHAP summary (beeswarm) plot in Fig-

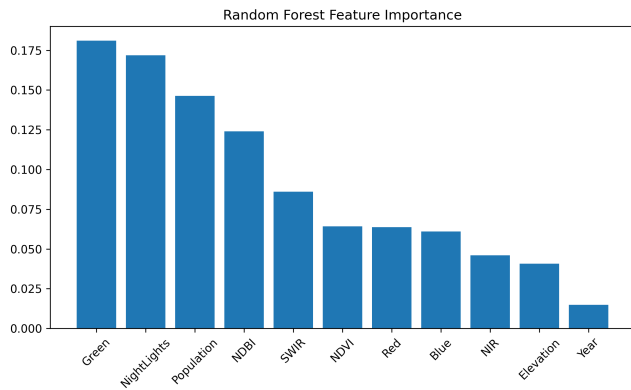


Figure 2: Random Forest feature importance showing that the Green spectral band, night-time lights, and population density are the most influential predictors of urban classification.

Figure 3 illustrates both the magnitude and direction of each predictor's contribution to urban classification. The results show that Night-time Lights produced the highest SHAP values, indicating that increasing light intensity strongly increases the probability of urban classification. Population Density also exhibited substantial positive contributions, reflecting the close association between densely populated areas and urban development. Conversely, vegetation-related predictors generally produced negative SHAP values, indicating that greater vegetation cover decreases the likelihood of urban classification. The SHAP analysis therefore provides an interpretable explanation of the model's decisions, demonstrating that predictions are driven by physically meaningful relationships rather than spurious correlations. While the Random Forest feature importance ranked the Green spectral band as the most influential predictor, SHAP identified Night-time Lights as the dominant contributor to individual predictions. This difference reflects the distinct objectives of the two methods: Random Forest importance measures a variable's contribution to reducing classification impurity, whereas SHAP quantifies each feature's contribution to individual model predictions. Together, these complementary analyses provide a more comprehensive understanding of the factors driving urban classification.

The results show that Night-time Lights produced the highest SHAP values, indicating that higher light intensity strongly increases the likelihood of a location being classified as urban. Population density also demonstrated a strong positive relationship with urban classification. Conversely, NDVI values exhibited negative SHAP contributions, suggesting that areas with higher vegetation density are more likely to be classified as non-urban.

The SHAP feature importance bar plot (Figure 4) confirms the dominant role of night-time lights and population density in the classification process. The dependence plot for night-time lights (Figure 5) further illustrates how increasing light intensity corresponds to higher SHAP values, reinforcing its strong association with urban development.

From an interpretability perspective, these results highlight the importance of combining predictive modeling with fea-

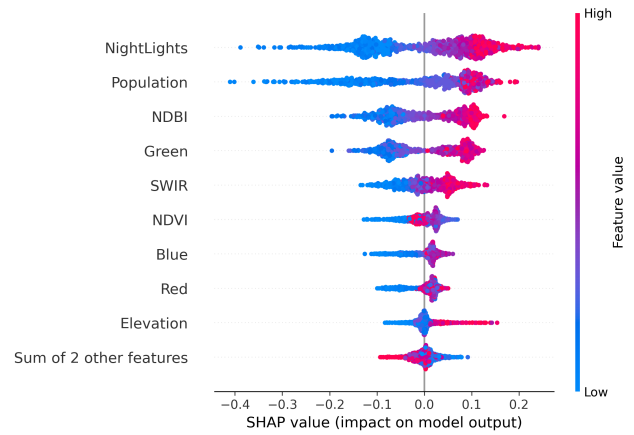


Figure 3: SHAP summary plot illustrating the global influence of predictor variables on urban classification, with night-time lights and population density showing strong positive contributions, while NDVI exhibits negative effects.

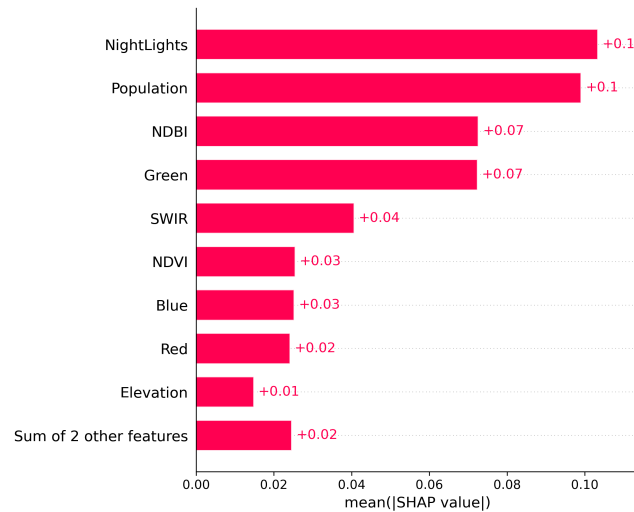


Figure 4: SHAP feature importance based on mean absolute SHAP values, confirming the dominant role of night-time lights and population density in urban classification.

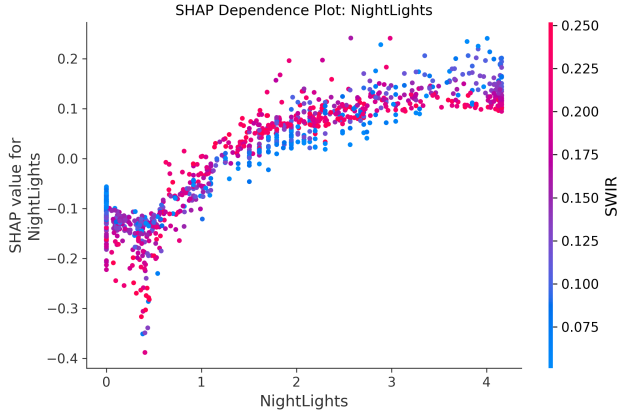


Figure 5: SHAP dependence plot for night-time lights showing a strong positive relationship between light intensity and the likelihood of urban classification.

ture attribution methods to uncover domain-relevant patterns. The consistency between SHAP-derived feature importance and known urban indicators suggests that explainable AI can provide reliable insights into complex geospatial processes.

3.4 Key Findings

Overall, the results demonstrate that all three models achieved consistently high performance in urban classification, with XGBoost showing a slight advantage in AUC. The strong discriminatory power of the multisource geospatial dataset highlights the complementary roles of spectral and socio-economic variables, with the Green spectral band emerging as the most influential predictor, followed by night-time lights and population density. This suggests that both surface reflectance characteristics and human activity indicators are critical for distinguishing urban areas. SHAP-based explainability further clarifies model behavior, revealing a negative association between vegetation (NDVI) and urban areas, and a positive contribution from built-up indices such as NDBI. Collectively, these findings emphasize the robustness and interpretability of the proposed approach, demonstrating its effectiveness for accurate and scalable urban mapping in rapidly growing cities.

3.5 Spatial Distribution of Urban Areas

The spatial distribution of predicted urban areas in Lagos for 2020 is shown in Figure 6. The map was generated using the XGBoost model, which achieved the highest classification performance among the evaluated models. The results reveal a clear concentration of urban areas in central Lagos, with expansion extending into surrounding peri-urban regions. The spatial pattern highlights the heterogeneous nature of urban development, with varying intensities of urbanization across the study area.

4 Discussion

The results demonstrate that machine learning and deep learning approaches can effectively classify urban and non-urban areas using multisource geospatial data. All three

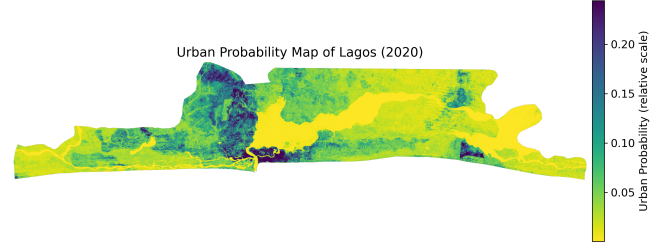


Figure 6: Urban probability map of Lagos (2020) generated using the proposed model. Higher values indicate a greater likelihood of urban presence. Probabilities are visualized using a relative scaling (2–98 percentile stretch) to enhance spatial contrast.

models: Random Forest, XGBoost, and the neural network, achieved similar performance, with an accuracy of 0.87 and AUC values above 0.94. These findings are consistent with previous studies showing that ensemble tree-based models perform strongly in remote sensing classification tasks due to their ability to model complex nonlinear relationships [Belgiu and Drăguț, 2016].

The comparable performance between the machine learning models and the neural network suggests that the selected predictor variables provide strong discriminatory power for urban detection. While deep learning models often outperform traditional approaches when spatial context or image-based features are used, tree-based ensemble methods remain highly competitive when working with tabular datasets derived from spectral and socio-economic variables [Du K-L, 2025].

These findings emphasize that, in tabular geospatial settings, the quality and diversity of input features may outweigh the choice of model architecture. This aligns with broader observations in machine learning that data representation is often the primary driver of model performance, particularly when heterogeneous data sources are effectively combined.

The explainable AI analysis provided valuable insights into the drivers of urban classification. Night-time lights emerged as the most influential predictor, highlighting the strong relationship between artificial lighting intensity and urban development. This aligns with previous studies where night-time lights serve as proxies for urbanization and economic activity. Population density also played a significant role, reinforcing the importance of integrating socio-economic data with remote sensing variables to improve classification performance [F. Zhao and Xie, 2021].

More broadly, the integration of explainable AI within geospatial modeling frameworks enables a shift from purely predictive systems to decision-support tools that provide actionable insights. This is particularly important in urban studies, where transparency and interpretability are essential for policy and planning applications.

Spectral indices such as NDBI contributed substantially to the classification results by effectively capturing built-up surface characteristics, while NDVI exhibited a negative relationship with urban classification due to its association with vegetation. These findings confirm the complementary roles of spectral and socio-economic variables in urban detection.

The prominence of the Green spectral band as the most influential predictor further highlights the importance of surface reflectance characteristics in urban classification. Unlike derived indices such as NDVI and NDBI, the Green band retains raw spectral information that captures subtle variations between built-up surfaces, vegetation, and transitional land cover types. In heterogeneous urban environments such as Lagos, where built-up areas are often interspersed with vegetation and bare soil, this band provides enhanced sensitivity to mixed land-use patterns. The high importance of the Green band therefore suggests that spectral reflectance differences play a critical role in distinguishing complex urban landscapes, complementing socio-economic indicators such as night-time lights and population density.

The SHAP analysis further demonstrated the value of explainable AI techniques in geospatial modeling by providing transparent insights into model behavior. This level of interpretability is particularly important for urban planning applications, where understanding the drivers of spatial patterns is as critical as achieving high predictive accuracy. These findings align with recent developments in explainable geospatial artificial intelligence, which emphasizes interpretable models as essential for trustworthy urban analytics and spatial decision support [Liu *et al.*, 2024]

The spatial prediction map provides additional insight into the distribution of urban areas across Lagos. The results reveal a pronounced concentration of urban development within the metropolitan core, with expansion extending into surrounding peri-urban regions. The probability-based representation highlights spatial gradients of urban intensity, indicating that urbanization is heterogeneous rather than uniform across the landscape. These spatial patterns reflect typical outward expansion in rapidly growing cities and underscore the importance of integrating spatial outputs with model evaluation metrics.

Overall, while the performance metrics confirm strong classification capability, the spatial outputs reveal the geographic structure and heterogeneity of urban growth. These findings demonstrate the added value of combining machine learning, multisource data, and explainable AI to achieve both accurate and interpretable urban mapping. Such integration aligns with the broader objective of developing trustworthy, transparent, and interpretable AI systems for remote sensing applications, where model explanations support confidence in geospatial decision-making [Wang *et al.*, 2024].

Despite the encouraging results, this study has some limitations. The proposed framework was evaluated using multisource geospatial data from Lagos, Nigeria, and its performance may vary in cities with different urban structures or data characteristics. Furthermore, the analysis relied on tabular representations of the predictor variables and did not explicitly model spatial neighbourhood relationships. Future research should evaluate the framework across diverse geographic settings and investigate spatially explicit deep learning approaches to further improve model generalizability.

5 Conclusion

This study presents an interpretable machine learning framework for urban classification using multisource geospatial data in Lagos, Nigeria. By integrating spectral, socio-economic, and temporal features, the proposed approach achieved strong predictive performance across multiple models, with Random Forest and XGBoost attaining comparable Accuracy, Precision, Recall, and F1-score values of 0.87, while all three models achieved AUC values above 0.94. These results demonstrate the effectiveness of multisource tabular data for accurate urban classification.

Beyond predictive performance, the incorporation of SHAP-based explainable AI provides valuable insights into the factors driving model predictions. Night-time Lights and Population Density emerged as dominant contributors to urban classification, while spectral variables provided complementary information for distinguishing complex urban environments. Together with conventional Random Forest feature importance analysis, SHAP enhanced the transparency and interpretability of the proposed framework by explaining both global feature importance and individual prediction behavior.

The proposed framework offers a scalable and practical tool for monitoring urban growth in rapidly developing regions. Building on these findings, future work will explore the integration of spatial context through convolutional and spatially explicit deep learning models and evaluate the framework across a broader range of cities and geospatial datasets. More broadly, this study demonstrates that combining heterogeneous data sources with interpretable machine learning provides a robust paradigm for transparent and trustworthy geospatial AI applications.

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